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Artificial Intelligence Design for Ship Structures: A Variant Multiple-Input Neural Network-Based Ship Resistance Prediction

In this work, we have developed a data-driven artificial intelligence (AI) solution to assist the ship hull design process. Specifically, we have developed and implemented an AI-based multiple-input neural network model to realize the real-time prediction of the total resistance of the ship hull structure while avoiding the inconsistent estimates from different types of design input parameters. It is demonstrated that the developed AI-based machine learning algorithm as a prediction tool can assist the ship hull design process by accurately providing the total resistance of ship hulls in real time. Moreover, we have conducted design tasks to validate the proposed method, and the validation results show that a well-trained artificial neural network model can avoid the problem of different sensitivities due to the different degrees of influence of the input parameters on the output parameter. The proposed AI-based data-driven solution provides a real-time hydrodynamic performance calculation, which can predict the hyperdynamic performances of ship hulls based on their geometry modification parameters. This approach gives a consistent prediction in terms of accuracy when facing different geometry modification parameters, and it in turn provides a fast and accurate AI-based method to assist ship hull design to achieve an optimum forecast accuracy in the entire design space, making an advance to artificial intelligence assist design in naval architecture engineering. [DOI: 10.1115/1.4053816]

Keywords: approximation-based optimal design, artificial intelligence, data-driven design, design theory and methodology, expert systems, machine learning, simulation-based design

1 Introduction

The ship design stage [1] is one of the most time-consuming and labor-intensive steps in shipbuilding engineering. The designer needs to comprehensively consider all the technical requirements and performance requirements of the ship, including but not limited to the overall arrangement, ship specifications, and benefits evaluation. A ship involves multiple conflicting performance criteria, making the ship design process even more complicated [2]. Since the ship design process is a multidisciplinary-cross mission, ship designers usually will divide the whole process into several substages based on design goals.

In ship engineering, the ship hull design belongs to the initial design, whose main objective is to design a ship hull that has the best hydrodynamic performances under design conditions. As shown in Fig. 1, in the initial design stage [3], designers need to finish the hull form design, and then, based on the selected hull form, designers need to determine the tank plan, capacities, intact, and damage stability of the hull form.

Due to the need to determine the ship hull with optimal hydrodynamic performances, the ship hull design process always repeats many design loops. Based on one or several initial hull models, designers need to repeat the design space exploration, geometry modification, and performance evaluation until an ideal hull form is obtained.

When we analyze the time and workload spent in the ship hull design stage, we can find that the design workload and time are mainly spent on geometry modification and performance evaluation. Specifically, after designers acquire a new ship hull based on the geometry modification, designers need to model the new hull form to enable it to be calculated by computational fluid dynamics (CFD) [4] software, and then wait for the software to finish the calculation process to get its hydrodynamic performances. Thus, if there exists a regression tool that can replace the traditional geometry modification process and performance evaluation process, then the design time and workload can be significantly shortened. No matter how many times the design loop needs to be operated, designers can obtain the performances of hull form after they design a new one in a short time.

In some recent studies, artificial intelligence (AI)-based artificial neural network (ANN) techniques [5] are proven to have an excellent performance to assist engineering design since the ability to extract the complicated relationship between input and output sampling data. For instance, Park et al. [6] utilized ANN techniques to achieve accurate and efficient aerostructure wing design optimization. Du et al. [7] proposed a fast, interactive design framework for airfoil aerodynamic optimization and achieved response predictions based on multiple ANN models. Zhang et al. [8] developed a hybrid method based on teaching–learning-based optimization and ANN technique to solve engineering optimization problems.

Moreover, Chen et al. [9] utilized convolutional neural network (CNN) to predict the aerodynamic coefficient of an airfoil; Li et al. [10] proposed a long short-term memory (LSTM) based

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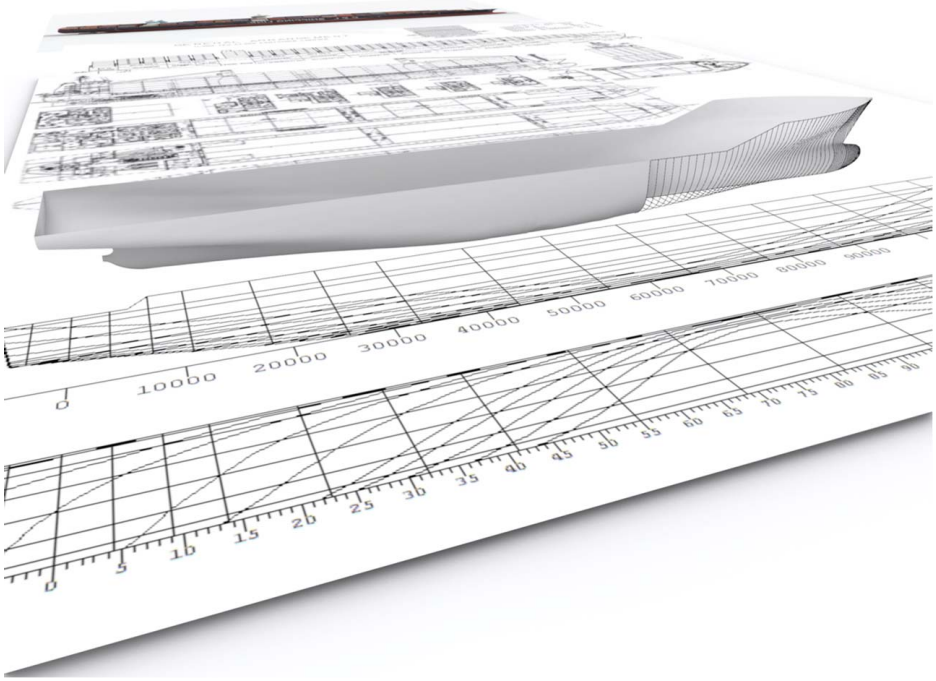


Fig. 1 The initial design stage of ocean and naval architectural engineering

interactive design framework to shorten the design optimization time, and Achour et al. [11] developed a generative adversarial network (GAN) network model for airfoil shape optimization. Some researchers have also used ANN techniques in ship engineering design assistance [12,13].

However, it is easy for it to become susceptible to specific input variables, which is common in its application. In the ship hull design, in order to be able to compare the performance differences between different ship hulls, the ANN model used to evaluate the hydrodynamic performance may not have very high accuracy, while it must be able to accurately predict how the hydrodynamic performance changes with changes in geometry modification parameters. Specifically, in the initial design stage, the prediction accuracy becomes not that important since a set of more accurate designs will be executed after this stage. Indeed, the more fit between the performance trend changed with inputs predicted by ANN models and the actual performance trend changed with inputs, the more practical the ANN model in the initial design stage. As the performance evaluation tool, an ANN that cannot deal with the situation in which design variables have different responses will mislead the optimization algorithm and make the optimization algorithm miss the optimal object.

In this work, we adopted a multiple-input neural network (MINN) model to predict and evaluate the total resistance for ship hulls based on geometry modification parameters, which provides an effective and practical method to evaluate the performance of hull forms under the premise of ensuring similar prediction sensitivity for different geometry modification parameters. Hence, the influence of different sensitivity of the inputs to the outputs to the final network model can be counteracted. The MINN model will be trained to simultaneously perform the data standardization function for different categories of parameters and the performance evaluation. This approach will advance the state-of-the-art artificial intelligence assist design (AIAD), which realizes real-time prediction of the hydrodynamic performance of ship hulls and seeks to assist the ship hull design. We believe that the AI technique can assist the ship hull design process, with traditional approaches incurring considerable time and workload due to the complicated geometry modification and performance evaluation processes. The application process of MINN in ship hull design is shown in Fig. 2.

In the rest of the paper, we begin with a brief introduction to the AI-based ANN technique and the MINN model used in this paper. Later, we briefly introduce the method of expressing a ship model to a set of geometry modification parameters. We then outline the process for collecting the sample dataset and the design space of the dataset. The detailed process about the hyperparameters tuning process and model training process was represented after demonstrating the establishment of the MINN model. Based on several validation examples, we seek to demonstrate that the MINN model proposed in this paper can accurately predict the total resistance of hull forms and ensure the MINN has similar sensitivity for different geometric deformation parameters. Last, an application case of the MINN model in hull optimization was followed to show its practicality.

2 Artificial Neural Network and MINN Algorithms

2.1 A Brief Introduction to Artificial Neural Network Algorithm. Artificial neural network (ANN) algorithm is a continually evolving field based on the thought of artificial intelligence (AI). ANN algorithm has a unique ability to find a mathematical model, or an algorithm, to extract internal relationships from complex or even ambiguous data.

The robust analysis ability of ANN algorithms comes from the exquisite and unique design of their architecture. The architecture of ANN algorithms refers to how the inner neurons are connected, which is an essential factor when determining the functions and learning process of ANN algorithms. In neural network engineering, the architecture design represents the mind of neural network engineers with their idea of what kind of function they want the ANN algorithms to achieve. The amount of data sample required to achieve sufficient prediction accuracy and the training time can be significantly reduced when selecting the right.

Due to the diversity of architecture, ANN algorithms now achieved very successful applications in many fields. However, in the field of artificial intelligence assist design (AIAD), especially for ship design-assist, the potential of the ANN algorithms still can be further explored.

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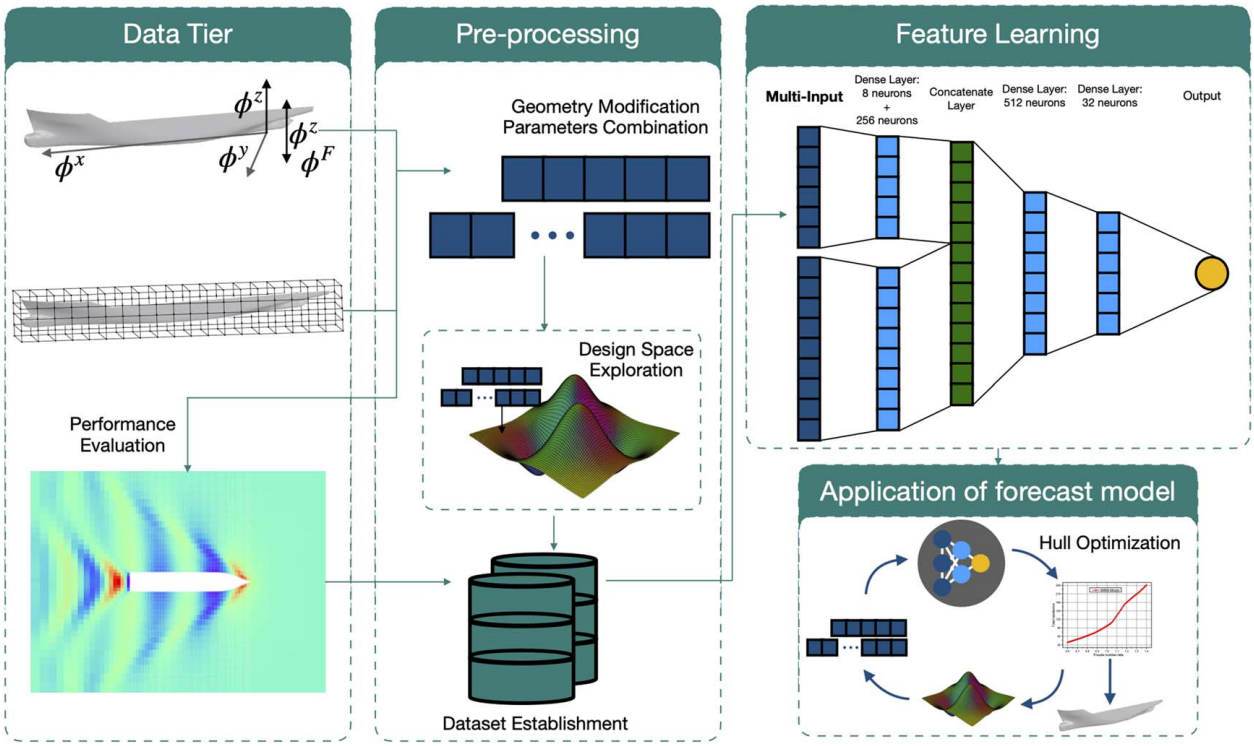


Fig. 2 Schematic diagram of the proposed AI-assisted design framework that uses machine learning neural network approach

2.2 Multiple-Input Neural Network. The ANN algorithm adopted in this paper is called the multiple-input neural network (MINN) [14,15], which was developed to deal with a task that contains many types of input parameters. In real life, ANN engineers sometimes may encounter a situation that needs to establish a neural network to process multiple-input data types, including quantity, classification, and image data. People often use the term "mixed data" in machine learning to indicate that a task contains multiple-input data types. Meanwhile, to facilitate the training of the ANN algorithm, ANN engineers always want to construct an end-to-end network, where the information signals can flow from the input to output without barriers or gaps.

Before the MINN architecture has been proposed, neural network engineering may need to build multiple network models to handle mixed data. A bad situation is to use one network model to deal with image information and use another network model to deal with digital information, which makes the training process very troublesome. MINN architecture enables neural network engineering to build an end-to-end network model to deal with mixed data tasks. An end-to-end network model making the information flow from the input to the output without a break enables a learning algorithm [16] to adjust all network model parameters without barriers.

The schematic diagram of the MINN algorithm is shown in Fig. 3. As shown in Fig. 3, before each data type is imported into the main network structure, a separate preprocessing step is performed, including scaling, normalization, and feature engineering.

3 A Method to Simplify the Hull Form to a Vector

3.1 Basic Idea for Simplification. At the beginning of the design process, designers usually will select an initial hull form as a prototype instead of starting to design a hull form from scratch, giving us a way to simplify the hull form to a vector, that is, convert the vector of the hull form to the vector of the geometry modification process of the hull form. Hence, the hull form can be represented by the initial hull model and the vector representing the geometry modification process. The geometry modification method adopted in this paper combined linear transformation and free-form deformation (FFD) [17], enabling us to represent the geometry modification process of the hull form into a set of geometry modification parameters.

The geometry modification process of the hull model can be divided into two steps: (1) the hull principal parameters modification based on linear transformation and (2) the hull shape geometry modification based on FFD technology. Although the hull model generated by the geometry modification method used in this work is highly dependent on the initial hull model, we can express the

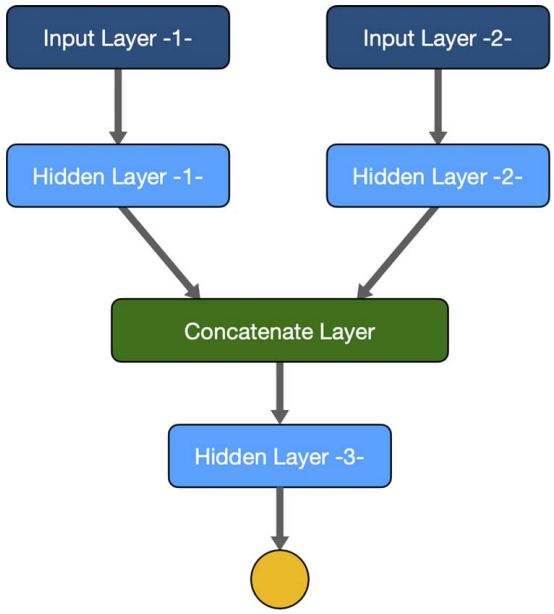


Fig. 3 Schematic diagram of the multi-input neural network model

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hull model with the geometry modification parameters through this method. More importantly, by increasing the number of control points and the range of parameters, the design space can be highly expanded until it satisfies the design requirements.

3.2 Geometry Modification of Principal Dimensions. In step 1, the principal dimensions of the ship hull model will be adjusted, and we called this step the geometry modification of principal dimensions. Since the geometry modification of the target object based on the FFD technique is actually directly acting on the target object point, it will bring great convenience when we transform the initial hull model into a matrix of discrete hull points. As shown below, we use a matrix P_o containing n hull points to represent the initial hull model:

$$\boldsymbol{P}_o = \begin{bmatrix} x_1 & y_1 & z_1 \\ & \vdots & \\ x_i & y_i & z_i \\ & \vdots & \\ x_n & y_n & z_n \end{bmatrix} \tag{1}$$

where (x_i, y_i, z_i) are the coordinate values of ith point.

Based on the initial matrix of discrete hull points, we can use four principal dimension modification parameters to perform the principal dimension modification and then obtain the intermediate matrix of discrete hull points for the geometry modification of the hull shape in the next step. As shown in Eq. (2), the principal dimension modification parameters, including length rate ϕ^x, breadth rate ϕ^y, height rate ϕ^z, and draft rate ϕ^d. The intermediate matrix of discrete hull points P_I is given as follows;

$$\boldsymbol{P}_I = \boldsymbol{P}_o \cdot \begin{bmatrix} \phi^x \\ \phi^y \\ \phi^z \end{bmatrix} + \phi^z \cdot \phi^d \cdot \begin{bmatrix} 0 & 0 & D_o \\ & \vdots & \\ 0 & 0 & D_o \\ & \vdots & \\ 0 & 0 & D_o \end{bmatrix} \tag{2}$$

where D_o is the draft of the initial model.

3.3 Geometry Modification of the Hull Shape. In step 2, the ship hull shape will be adjusted. The parallelepiped lattice box can be determined based on the origin position and size of the lattice box. In this work, we determine the origin position and size of the lattice box based on the extreme value of the matrix of discrete hull points, which ensures that the lattice box could enclose the entire hull model.

Suppose that the maximum coordinate values of the points in the matrix P_o on the X-axis, Y-axis, and Z-axis are $\hat{x}_{\max}$, $\hat{y}_{\max}$, and $\hat{z}_{\max}$, and the minimum coordinate values of the points in the matrix P_o on the X-axis, Y-axis, and Z-axis are $\hat{x}_{\min}$, $\hat{y}_{\min}$, and $\hat{z}_{\min}$, respectively, then based on the correction parameter λ, the lattice length B^l, width B^b, and height B^h of the lattice box are given in Eq. (3). Note that the notation $\odot$ in the above equation denotes the element multiplication.

$$\begin{bmatrix} B^l \\ B^b \\ B^h \end{bmatrix} = \begin{bmatrix} \hat{x}_{\max} - \hat{x}_{\min} \\ \hat{y}_{\max} - \hat{y}_{\min} \\ \hat{z}_{\max} - \hat{z}_{\min} \end{bmatrix} \odot \begin{bmatrix} \phi^x(1+2\lambda) \\ \phi^y(1+2\lambda) \\ \phi^z(1+2\lambda) \end{bmatrix} \tag{3}$$

Once the lattice box and control points are determined, we can obtain the final matrix of discrete hull points based on the current position of all control points. Assuming that the set of the current position of all control points is $\bar{\boldsymbol{v}}$, then the deformed hull points matrix P_F can be obtained based on the FFD calculation, which follows Eq. (4):

$$\boldsymbol{P}_F = \mathrm{FFD}\langle \boldsymbol{P}_I, \boldsymbol{B}, \bar{\boldsymbol{v}} \rangle \tag{4}$$

The FFD technique used in this paper is based on the PyGem python library [18]. With the help of the FFD technique, we can use the position of control points to express the geometry modification of the hull surface shape.

4 Dataset Establishment

4.1 Selection of the Initial Hull Model. To train an ANN algorithm, we need to produce enough hull models to establish a dataset based on the initial ship hull model, making the final trained ANN algorithm predict the performances of the hull model modified based on the initial ship hull model.

In this paper, we choose to develop an ANN algorithm to assist the design of traditional ships, such as bulk carriers and container ships, which are the most common ship types and often need to be redesigned to meet requirements. The initial ship hull model adopted in this paper is the KRISO container ship (KCS) [19]. For the accuracy verification convenience of the calculation method, the KCS model adopted in this work is a 1:31.6 scale model [20].

4.2 Performance Evaluation Process. Since the ANNs can be trained by virtual reality data [21], we adopted virtual computing as the performance evaluation method to generate data samples. To save the dataset establishment cost, we choose to use the potential flow method to evaluate the total resistance of the hull models.

According to the simplification guideline provided by the International Towing Tank Conference (ITTC), we divide the total resistance of a ship when sailing in still water into frictional resistance and wave-making resistance, where the frictional resistance is calculated based on the classic ITTC frictional resistance formula [22] and the wave-making resistance is calculated based on Dawson's method [23,24].

4.3 Design Space Exploration. As engineers design for an inherently uncertain world [25], design space exploration is essential. In this work, we preset the number of control points or the scope range of all geometry modification parameters to fix the design space. It is noticeable that we expand the design space by adding speed parameters to the sample to assist the initial design stage further. We used the Froude number [26], a dimensionless number, to describe the speed of all hull models. The relationship between the sample Froude number f with speed rate ϕ^f and design Froude number f_d is shown in Eq. (5).

$$f = f_d \cdot \phi^f \tag{5}$$

The scope ranges of the principal dimensions modification parameters are shown in Table 1.

Before we set the scope range of parameters used to modify the shape of the hull surface, we need to set up the lattice volume to enclose all parts of the initial hull model, which is shown in

Table 1 The scope range of the geometry modification parameters related to principal dimensions

Input parameter	ϕ^x	ϕ^y	ϕ^z	ϕ^d	ϕ^f
Variable range	[0.8, 1.2]	[0.8, 1.2]	[0.8, 1.2]	[−0.4, 0.4]	[0.7, 1.3]

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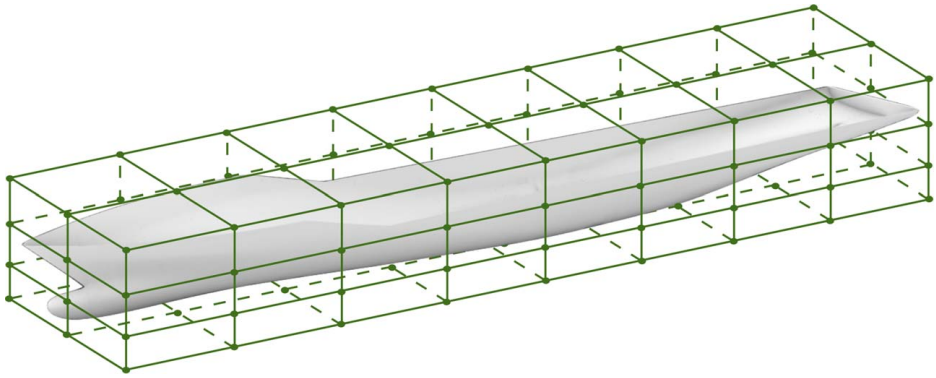


Fig. 4 Schematic diagram of a (9 × 3 × 4) finite-size lattice volume for a ship hull design case

Fig. 4. In this work, we created a (20 × 4 × 6) finite-size lattice volume of control points to enclose the hull model. The correction parameter λ used to gap the hull model with lattice boundary was set as 0.05 in this paper.

We set the range of control points according to the area where they are located to ensure that the control points located in the complex area have a larger range. The regional division of the hull form is shown in Fig. 5, and the specific location of different regions and the movement range of control points in the region are shown in Table 2. It is notable that some regions shown in Table 2 overlap each other so that a control point may be located under multiple areas. For one control point to share multiple regions, we always select the wider scope range for this point.

Moreover, as shown in Table 3 and Fig. 6, we preset the direction in which the control point can move, making the geometry modification process more intuitive and engineering-oriented. In detail, we force the control points located within boundary-1 near the parallel intermediate to move only in the y and z directions. The control points in boundary-2 and boundary-3 near the superstructure and the propeller can only move in the y direction.

In this work, the number of geometric modification parameters related to the main dimensions is 5. The final number of available geometric modification parameters related to control points is 496. Therefore, the design space is composed of 501 parameters.

4.4 Sample Acquisition Loop. Based on the initial model and the design space determined in the above sections, we can construct the dataset based on the sample acquisition loop while ensuring that all samples in the dataset are in our requirement area. In order for the final ANN model to be consistent in predicting hull performance in the design space, each sample in the database used to train the ANN model will be randomly selected.

In each circle of the sample acquisition loop, we randomly change the geometry modification parameters in the design space to get a new hull sample. Based on the randomly generated hull sample, a performance evaluation process will be executed to get the total resistance of this hull sample. Finally, the geometry modification parameters will be stored as the input of this hull sample, and the total resistance will be stored as the output of this hull sample. The sample acquisition loop will continue to execute until the data in the dataset reaches the number we set. At last, we constructed a dataset that contains 38,333 samples.

5 Construction and Training of the MINN Algorithm

5.1 Construction of the MINN Algorithm. Considering the situation that we use the fully connected neural network (FCNN) as the neural network architecture to predict the performance of hull forms, we need to combine all geometric modification parameters to the input vector of an FCNN. Furthermore, the FCNN architecture characteristics ensure that each input neuron can influence all neurons in the rest layers.

In other words, by mainly focusing on the principal dimension modification parameters, the learning algorithm can significantly decrease the loss value of the FCNN model. As a result, the prediction accuracy of the FCNN model will be very high when mainly the principal dimension modification parameters have been changed, and the prediction accuracy of the FCNN model will be relatively low when mainly the hull surface modification parameters have been changed. In detail, the loss function of the ANN model, as shown in Eq. (6), should be related to the principle dimension modification parameters $\boldsymbol{x}_p$ and hull surface modification parameters $\boldsymbol{x}_s$.

$$\boldsymbol{L} = L(\boldsymbol{x}_p, \boldsymbol{x}_s) \tag{6}$$

Since the geometry modification parameters related to principal dimensions have a more considerable influence on the loss function

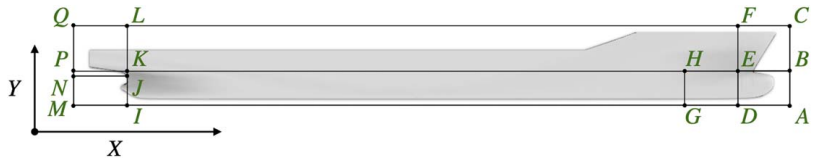


Fig. 5 Area division depends on the movement range of the control points

Table 2 Location of different areas and the movement range of the control points in the area

Area	ABED	ACQM	ABPM	KLQP	ABHG	ABED	ABHG	ABED
Direction	X	Y	Y	Y	Y	Y	Z	Z
Scope	[− 0.1, 0.1]	[− 0.1, 0.1]	[− 0.2, 0.2]	[− 0.3, 0.3]	[− 0.3, 0.3]	[− 0.4, 0.4]	[− 0.2, 0.2]	[− 0.3, 0.3]

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Table 3 Location of different areas and their fixed directions

Boundary area	Fixed coordinate
Boundary-1	X
Boundary-2	X, Z
Boundary-3	X, Z

than the geometry modification parameters related to the hull surface, it will account for a higher proportion of the loss function gradient, which guides the update direction of the inner parameters of the FCNN algorithm. During the training process, the learning algorithm will update the inner parameters of the FCNN model, which may prioritize the accuracy of principal dimension modification parameters, thereby ignoring the hull surface modification parameters to some extent. The different sensitivity of the FCNN model when facing different hull models may cause the design mistake when using the FCNN algorithm to assist the ship hull design.

We draw support from the multiple-input neural network (MINN) and wish to employ it to represent and process the structure characteristic information or data when we separately and simultaneously preprocess the two different kinds of geometric modification parameters while ensuring an end-to-end network. As shown in Fig. 7, the final neural network model consists of three parts. Part 1 and Part 2 are two FCNN models used to preprocess two different kinds of deformation parameters. Part 3 is the FCNN model used to receive two different intermediate information and obtain the final result. In the MINN model, several separate dense layers will preprocess the information of two categories of input parameters before they combine to the same layer. Hence, the separate dense layers will be trained to pro-process or standardize the different categories of input parameters to ensure that the final MINN is well performed after the tuning and training processes.

The construction, tuning, and training of the MINN algorithms are based on the Keras framework [27] with the TensorFlow backend [28]. All the construction, tuning, training, and validating processes were conducted based on the Adam optimizer [16].

5.2 Tuning Process of the MINN Algorithm. For tuning and training the MINN model, we construct a dataset that contains 38,333 samples and divide the dataset into a training dataset and testing dataset based on a ratio of 3:1.

The loss function is an index to evaluate the degree of completion of ANN model training. The lower the loss function value, the higher the degree of completion of deep learning model training. This paper uses the average absolute percentage error (MAPE) loss function [29] to direct the MINN model to narrow the gap between predicted and actual results.

Moreover, for easy comparison, we choose the mean and standard deviation of the absolute percentage difference of the samples in the test dataset as indicators to determine the prediction accuracy between different models. Assuming the total sample number of samples in the testing dataset is n, and the absolute percentage difference of the ith sample in the test dataset is p_i, then two indicators used to compare different MINN algorithms are shown as follows:

$$E_{\text{mean}} = \frac{\sum_{i=1}^{n_v} p_i}{n_v} \tag{7}$$

$$\text{and} \quad E_{\text{std}} = \sqrt{\frac{\sum_{i=1}^{n_v} (p_i - \mu)^2}{n_v}} \tag{8}$$

where μ is the population means.

Considering that we have selected the FCNN model as the unit, we only need to determine the number of hidden layers and neurons



Fig. 6 Area division depends on different fixed directions

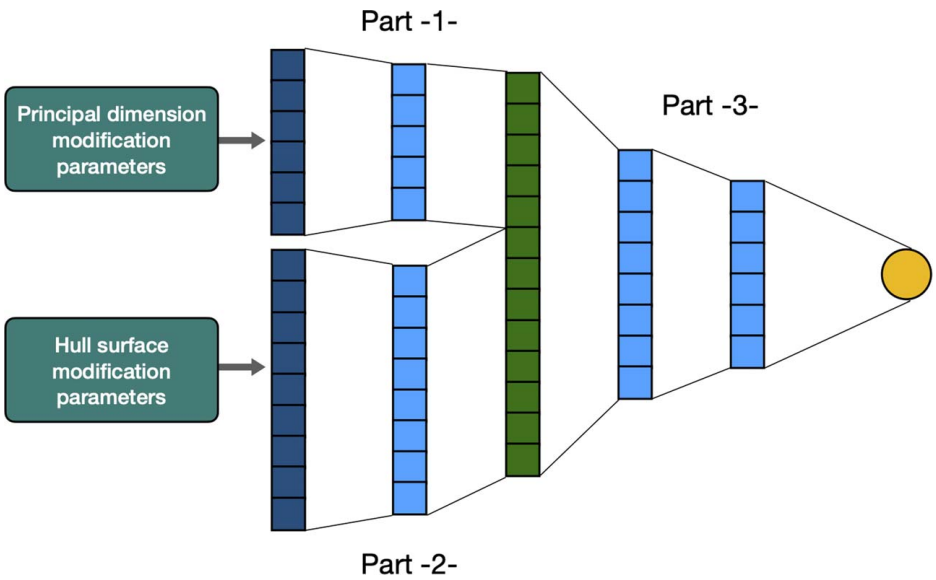


Fig. 7 Schematic diagram of the multi-input neural network model

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Table 4 Parts of tuning result of regularization parameters

	L1 rate	L2 rate	Dropout rate	E_{mean} (%)	E_{std} (%)
Case 1	0	0	0	21.381	18.385
Case 2	0.064	0	0	4.402	5.106
Case 3	0.001	0	0.001	7.596	6.309
Case 4	0	0.004	0.001	11.5217	9.063
Case 5	0.032	0.004	0.001	4.126	4.557

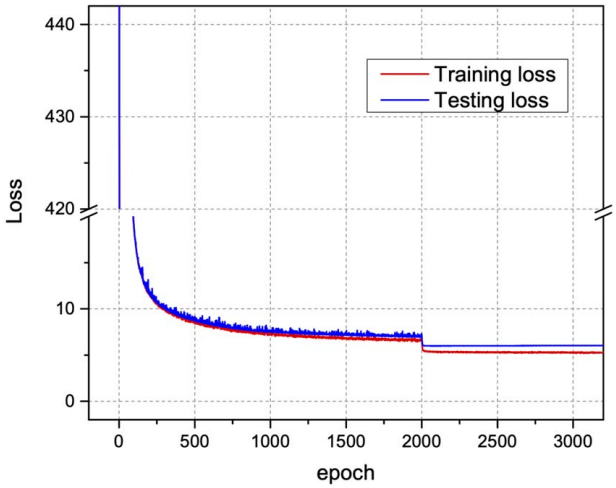


Fig. 8 Detailed structure of the multiple-input neural network model

in each hidden layer for each FCNN model. After trying and comparing 218 hidden layer cases, we choose the hidden layer structure of part 3 with low training loss, no obvious overfitting, and good prediction accuracy, which is (512, 32), to do the tuning of regularization parameters. Table 4 shows parts of tuning results with different regularization parameter combinations.

We choose the regularization parameter combination (case 5) with the lowest E_{mean} and E_{std} as the final choice. Meanwhile, based on *TensorFlow*, we can get the detailed structure of the final MINN model, which is shown in Fig. 8.

Under the regularization parameter combination of case 5, we continue the training process till epoch 3250 and stop since the loss value does not decrease after 2500, whose loss history is shown in Fig. 9. It is notable that there is a drop at epoch 2000, which is caused by the decrease of the learning rate. Finally, the parameters of the MINN model in epoch 3000 can be stored as a mathematical model to forecast the total resistance of any hull model in the design space. It has the lowest testing loss in the whole training process and shows no severe overfitting performance exists. In epoch 3000, the training loss decreased to 3.23, the validation loss decreased to 6.026, acquiring an outstanding forecast accuracy with E_{mean} of 3.946% and E_{std} of 4.760%.

6 Numerical Experiment

6.1 Forecast Accuracy Verification. We performed several validation cases by comparing the result evaluated based on the potential flow method and the ANN model to validate the forecast accuracy of the MINN model we developed in this work. Since two categories of geometry modification parameters represented all hull models in the design space, we validate the forecast accuracy of the MINN model in two steps.

In step 1, we only validated the geometry modification parameters related to the principal dimensions, and the geometry modification parameters related to the hull surface were fixed. To avoid the cases in the verification process that happen to be consistent with the samples that exist in the dataset, we fixed all the 496 geometry modification parameters related to the hull surface to 0.01. Based on these fixed modification parameters, we can validate the forecast accuracy of the MINN model when facing the change of geometry modification parameters related to principal dimensions by several cases, whose details are shown in Table 5. Note that the scope ranges of geometry modification parameters related to principal dimensions exist out domain region, which can further check the predictive power of the MINN model.

We verify each sample within the preset range at an interval of 0.01, and the final results can be seen in Fig. 10. Figure 10(*a*) shows the curves for total resistance predicted by the potential flow method and the MINN model when the length change rate is in the scope range (0.7, 1.3) with the domain range of (0.8, 1.2).

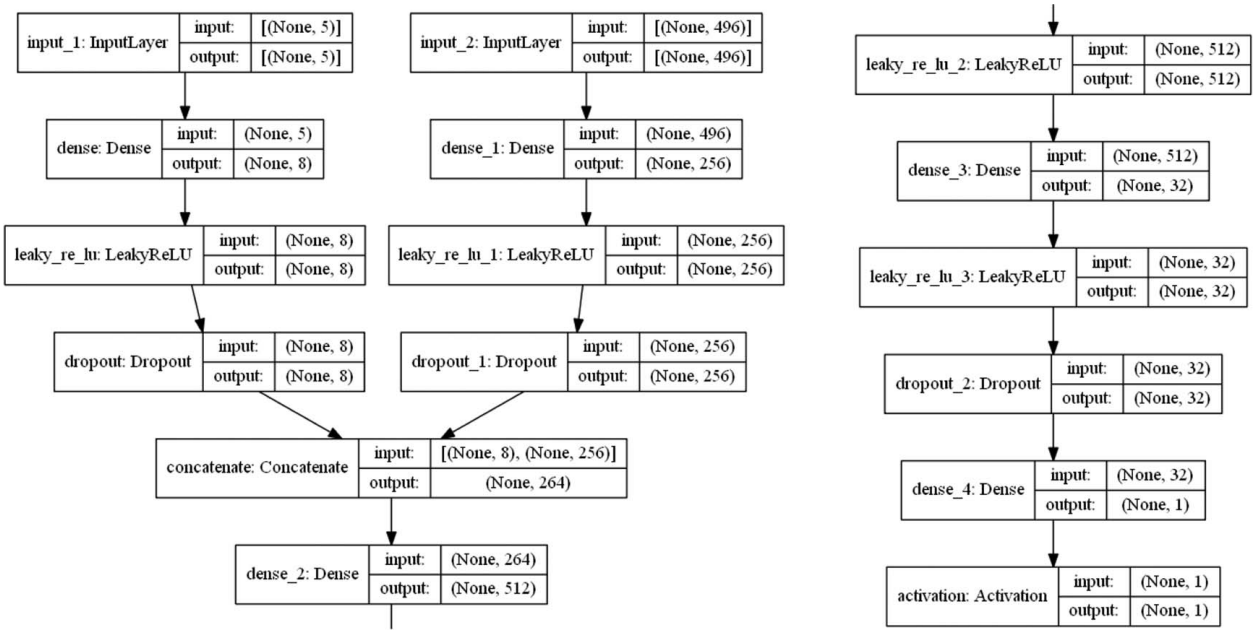


Fig. 9 The training history of the final MINN model

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Table 5 The details of the verification cases in step 1

	ϕ^x	ϕ^y	ϕ^z	ϕ^d	ϕ^f
Case 1	(0.7, 1.3)	1	1	0	1
Case 2	1	(0.7, 1.3)	1	0	1
Case 3	1	1	(0.7, 1.3)	0	1
Case 4	1	1	1	(−0.5, 0.5)	1
Case 5	1	1	1	0	(0.6, 1.4)

Table 6 The details of the verification cases in step 2

	ϕ^x	ϕ^y	ϕ^z	ϕ^d	ϕ^f	m_1	m_{496}
Case 6	1	1	1	0	1	(−0.1, 0.1)	0.1
Case 7	1	1	1	0	1	(−0.1, 0.1)	0.05
Case 8	1	1	1	0	1	(−0.1, 0.1)	0

It is worth noting that the potential flow method used in this work becomes instability and even error when the length rate is less than 0.76 due to the stubby hull model. Figure 10(*b*) shows the curves for total resistance predicted by the potential flow method and the MINN model when the breadth rate is in the scope range (0.7, 1.3) with the domain range of (0.8, 1.2). Figure 10(*c*) shows the curves when the breadth rate is in the scope range (0.7, 1.3) with the domain range of (0.8, 1.2). Figure 10(*c*) shows the curves when the height rate is in the scope range (0.7, 1.3) with the domain range of (0.8, 1.2). Figure 10(*d*) shows the curves when the draft rate is in the scope range (−0.5, 0.1) with the domain range of (−0.4, 0). Figure 10(*e*) shows the curves when the Froude number rate is in the scope range (0.6, 1.4) with the domain range of (0.7, 1.3). When we notice the domain area, we can find that the MINN model shows a significant ability to forecast the trend of the total resistance with the change of the principal dimensions modification parameters, where the difference of each test sample is very similar.

In step 2, we validate the geometry modification parameters related to the hull surface, and the geometry modification parameters related to the principal dimensions were fixed. Considering the number of control points, changing a single control point will not significantly impact the performance of the hull model. Hence, we generated the parameters related to the hull surface of test cases based on linear functions. All 496 hull surface parameters are given in the following equation:

$$m_i = m_1 + \frac{(m_{496} - m_1)}{495}(i - 1) \tag{9}$$

where m_1 is the movement amount of the first control point, m_{496} is the movement amount of the last control point, and m_i is the movement amount of the ith control point.

Hence, once we determine the value of m_1 and m_{496}, we can get the value of all rest parameters. We can also ensure that the test sample generated based on Eq. (9) is not the same as any sample in the dataset. Based on Eq. (9), we can validate the forecast accuracy of the MINN model when facing the change of geometry

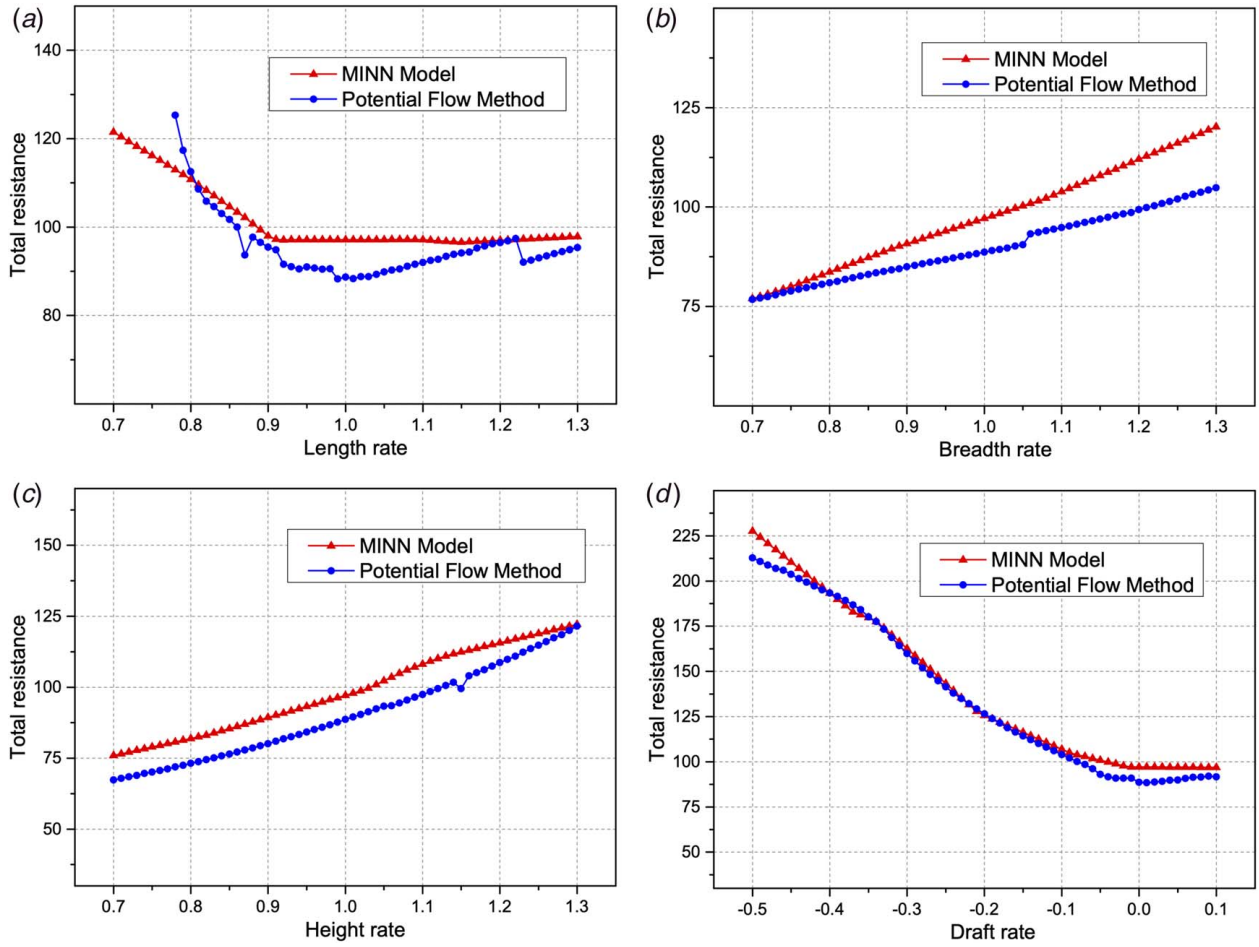


Fig. 10 (*a*) Curve of total resistance versus length rate, (*b*) curve of total resistance versus breadth rate, (*c*) curve of total resistance versus height rate, (*d*) curve of total resistance versus draft rate, and (*e*) curve of total resistance versus Froude number rate

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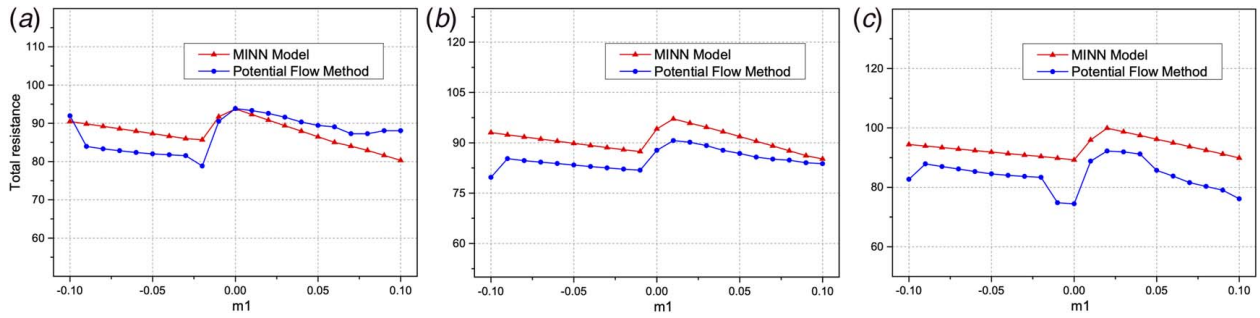


Fig. 11 (*a*) Verification curve of case 6, (*b*) verification curve of case 7, and (c) verification curve of case 8

Table 7 Performance metrics of the MINN model

Testing case	MAE	RMSE	MSE	MAPE	sMAPE
Case 1	4.69	5.34	28.5	5.10%	4.94%
Case 2	8.04	8.47	71.75	8.04%	9.4%
Case 3	9.22	9.27	85.99	10.46%	9.93%
Case 4	2.87	3.69	13.65	2.49%	2.55%
Case 5	7.39	7.56	57.17	8.09%	7.75%
Case 6	3.98	4.51	20.3	4.67%	4.64%
Case 7	5.76	6.2	38.47	6.81%	6.55%
Case 8	9.35	9.82	96.45	11.36%	10.67%

modification parameters related to the hull surface by several cases, whose details are shown in Table 6.

On the premise of ensuring the rationality of the hull model, we fixed m_1 in the range of −0.1 to 0.1 and changed the value of m_{406}. The final results can be seen in Fig. 11. For all three cases in Fig. 11, m_1 ranged from $(-0.1, 0.1)$ with an interval of 0.01. Figure 11(*a*) shows curves when m_{496} is fixed to 0.1, Fig. 11(*b*) shows curves when m_{496} is fixed to 0.05, and Fig. 11(*c*) shows curves when m_{496} is fixed to 0. In all 63 test samples in step 2, we found that the MINN model predicted the trend of the total resistance with the change of m_1 accurately.

It is worth noting that the error between the two methods increased when m_1 reached the boundary, which was caused by the decrease of the sample density. Moreover, a more representative performance of the MINN model, regarding the performance metrics mean absolute error (MAE), root-mean-squared error (RMSE), mean squared error (MSE), mean absolute percentage error (MAPE), and symmetric mean absolute percentage error (SMAPE) [30,31], are shown in Table 7. It is worth noting that the performance metrics in Table 7 are all measured based on the testing data samples in the domain range.

Based on two steps of forecast accuracy verification, we can find that the MINN model can be an efficient method to assist the ship design since it can predict the trend of hull performance with geometry modification parameters with high accuracy.

6.2 Sensitivity Similarity Verification. We continue to verify the sensitivity similarity of the MINN model by comparing its forecast accuracy with the FCNN model, which has the same hidden

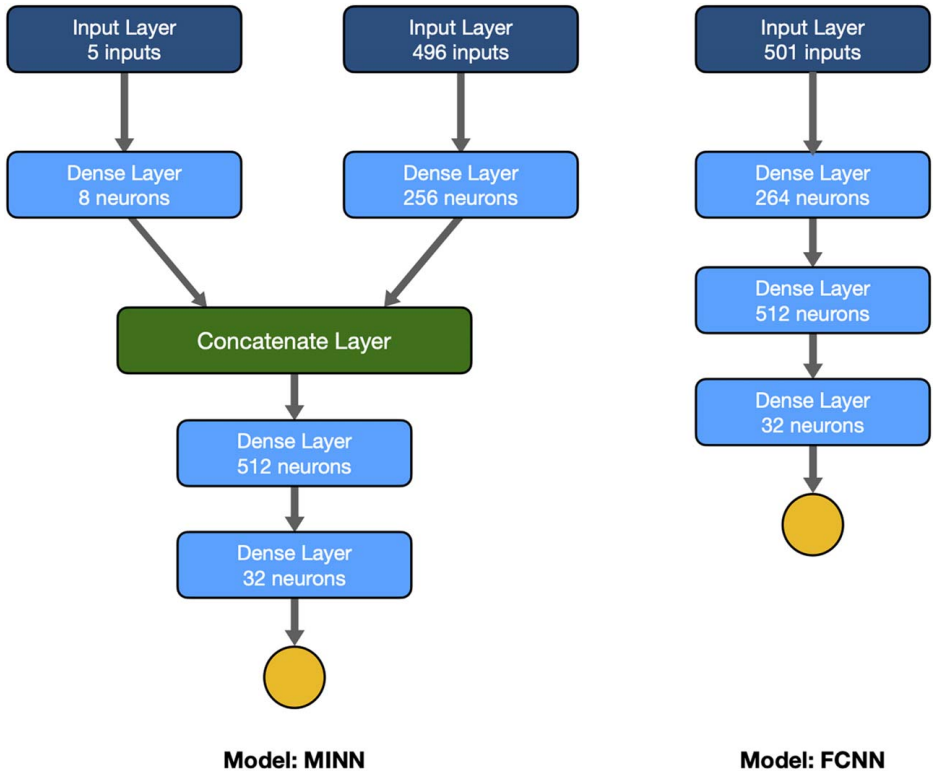


Fig. 12 Schematic diagrams of both the MINN model and the FCNN model

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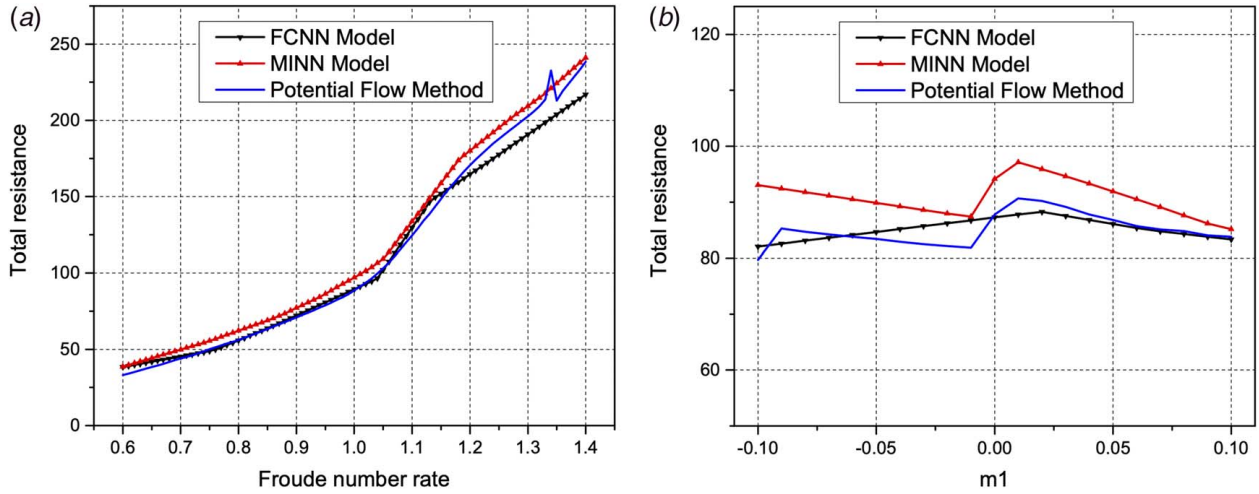


Fig. 13 (*a*) Verification curve of case 5 and (*b*) verification curve of case 7

Table 8 Performance metrics of the MINN model and the FCNN model

Model	Case	MAE	RMSE	MSE	MAPE	sMAPE
MINN	Case 5	7.39	7.56	57.17	8.09%	7.75%
FCNN		3.23	4.76	22.67	2.46%	2.47%
MINN	Case 7	5.76	6.20	38.47	6.81%	6.55%
FCNN		1.61	2.08	4.31	1.91%	1.89%

structure and hyperparameters. Schematic diagrams of both the MINN model and the FCNN model are shown in Fig. 12.

We trained the FCNN model to ensure it arrived at the optimal position based on the same hyperparameters of the MINN model. We stored the FCNN model located at epoch 5000 to do further verification.

We compare the forecast trend of the MINN model and the FCNN model with the actual trend to judge whether the forecast sensitivity of the two models is similar when facing different geometry modification parameters. Based on the parameters of case 5 and case 7, we verified each sample within the preset range at an interval of 0.01, and the final results can be seen in Fig. 13.

Figure 13(*a*) shows the curve for total resistance predicted by three methods when the Froude number rate is in the scope range (0.6, 1.4) with the domain range of (0.7, 1.3). Figure 13(*b*) shows the curve for total resistance predicted by three methods when the m_1 is in the scope range $(-0.1, 0.1)$ with the m_{496} fixed to 0.05. As shown in Fig. 13, we found that the trend curve predicted by the MINN model is closer to the actual trend than the trend curve predicted by the FCNN model regardless of the principal dimension parameters or the hull surface parameters. It is worth noting that when facing the geometry modification parameters related to the principal dimensions, no matter the MINN model or the FCNN model or both models show excellent forecast accuracy. However, when facing the geometry modification parameters related to the hull surface, the trend forecast accuracy of the FCNN model will be lower than the MINN model, where the MINN model predicts a curve consistent with the actual trend change.

Table 8 presents the performance of the MINN model and the FCNN model regarding the performance metrics MAE, RMSE, MSE, MAPE, and sMAPE. From the performance metrics shown in Table 8, we can find that the FCNN model shows a greater

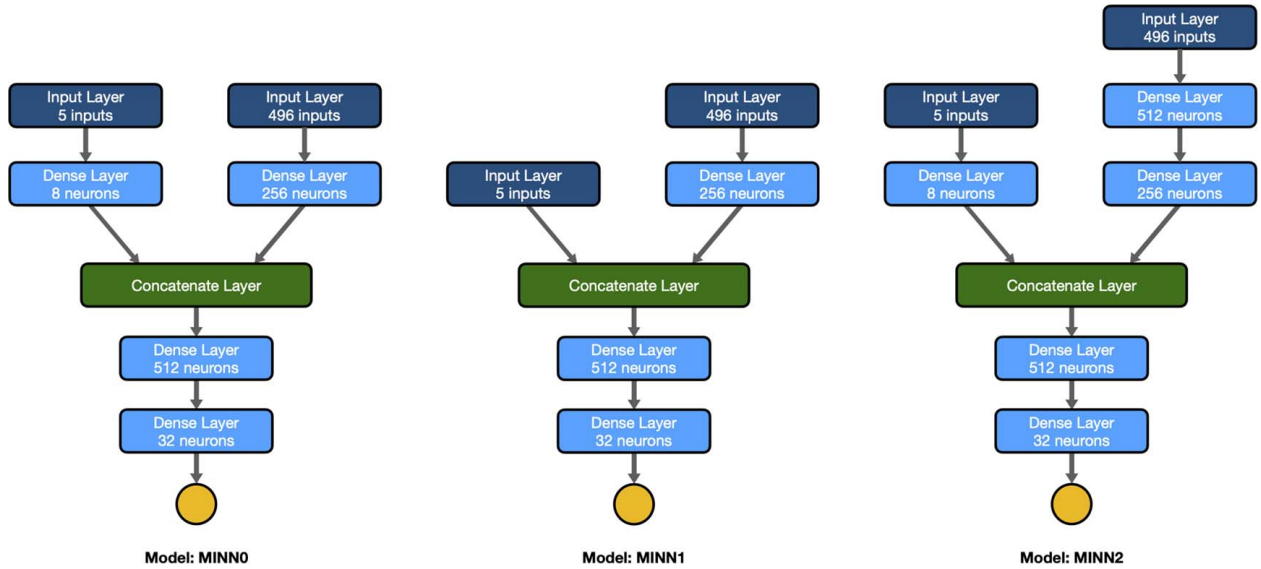


Fig. 14 Schematic diagrams of three MINN models with different layer numbers in branches

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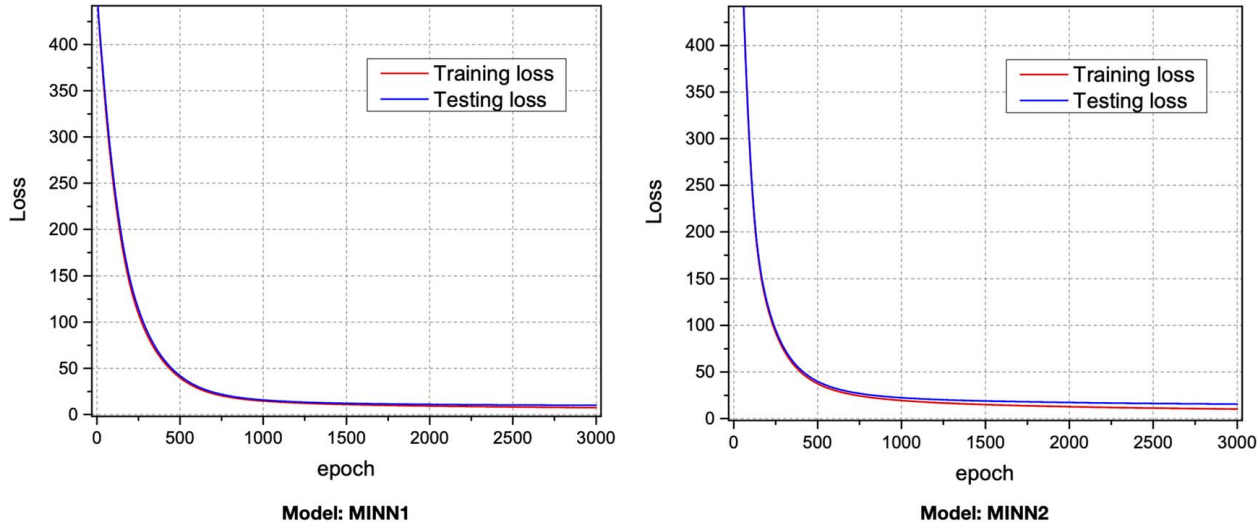


Fig. 15 The training history of the MINN1 and MINN2 models

Table 9 Performance metrics of three different MINN models

Model	MAE	RMSE	MSE	MAPE	sMAPE
MINN0	5.24	8.42	70.92	3.96%	3.92%
MINN1	6.81	10.53	110.98	4.95%	4.92%
MINN2	10.20	14.97	22421	7.30%	7.25%

prediction accuracy than the MINN model for case 5 and case 7. However, a more representative visualization of trend prediction performance is presented in Fig. 13. From Fig. 13, we can find that rather than the FCNN model shows a lower prediction error of two cases, the MINN model can more accurately predict the trend of hull performance with changes in parameters, which is a more critical function since ship designers need to know the trend to decide the optimal hull.

Hence, from all 102 test samples in Fig. 13, we concluded that the architecture of the MINN could effectively solve the problem of different sensitivity caused by the different degrees of influence of different categories of geometry modification parameters on the hull performance. The better trend performance of the MINN model is original from its structure. The structure design of the MINN model offers it the ability to get a better trend. We used numerical experiments to prove the MINN model and showed a better trend consistency than FCNN when using geometry modification parameters as input and the total resistance as output. However, the better consistency of the MINN model in other fields still needs to be further proved.

6.3 Analysis of the Influence of the Layer Number of Branches. As shown in Fig. 8, the MINN model has two branches, where one branch was used to preprocess the principal dimension modification parameters, and another branch was used to preprocess the hull surface modification parameters. In order to verify the influence of the preprocessing process on the predictive ability of the MINN model, we constructed and trained two different MINN models which have different layer numbers in each branch in comparison with the final adopted MINN model. Schematic diagrams of three MINN models with different layer numbers in branches are shown in Fig. 14.

From Fig. 14, one can find that all three MINN models have different layer numbers in branches, representing the degree of preprocessing to different types of input. If taking the MINN0 model as the standard model, the MINN1 model directly imports the principal dimension modification parameters into concatenate layer without

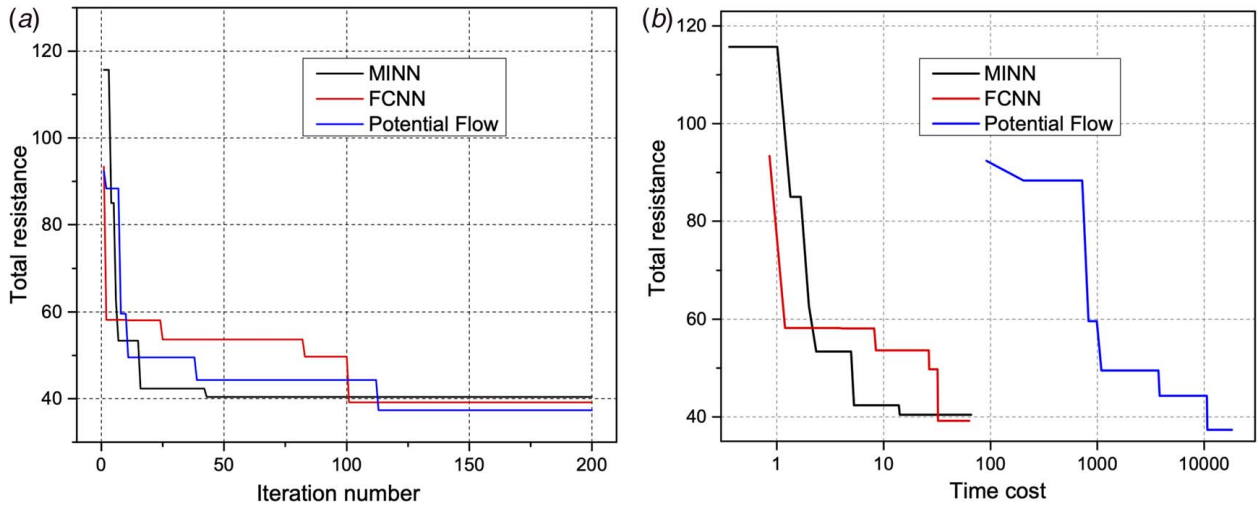


Fig. 16 (*a*) Curve of total resistance versus iteration number and (*b*) curve of total resistance versus time cost

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any preprocessing. Moreover, the MINN2 model adopts two layers to preprocess the hull surface modification parameters. To analyze the impact of layer number of branches on the MINN's performance, the regularization parameters of the MINN1 and MINN2 models are set to the same hyperparameters as the MINN0 model.

The training history of the MINN1 and MINN2 models are shown in Fig. 15. After training the MINN1 model and MINN2 model, we employ MAE, RMSE, MSE, MAPE, and sMAPE metrics to evaluate the three models predictive performance. Table 9 presents the metrics of all three models' when using the same testing dataset. As shown in Table 9, the degree of preprocessing will highly affect the predictive ability of the MINN model. The MINN0 model demonstrates the most robust predictive ability, even though it does not mean the structure of the MINN1 and MINN2 models are not suitable for dealing with inputs with a different degree of influence. By changing the regularization parameters or increasing data samples, both MINN1 and MINN2 models might perform better in specific tasks.

6.4 The Application of the MINN Model in Hull Optimization. Based on the forecast accuracy and sensitivity similarity verification, the MINN model can be considered a powerful tool for real-time prediction of the total resistance of any hull model in the design space. This section will perform a hull design case based on the FCNN model, the MINN model, and the potential flow method. Meanwhile, the design cost and results will be compared to show the advantages and disadvantages of the three methods.

In the design process, we can either use the FCNN model, or the MINN model, or the potential flow method as the performance evaluation simulation method to conduct the simulation-based design process. In the actual hull design process, the engineer must consider all the hydrodynamic metrics comprehensively. However, since we only consider the total resistance of the hull form, the total resistance will be the only metric to evaluate the performance of the hull form and its design method here, which transforms the hull design process into a hull shape optimization process. Hence, we can treat the geometric modification parameters as the input and treat the total resistance as the output, finding the hull form with the smallest total resistance in the design space based on the optimization algorithm.

For the sole purpose of comparing the different design processes among the FCNN model, the MINN model, and the potential flow method, the optimization algorithm used here should be simple and easy to operate. Therefore, we choose the random search algorithm [32] as the optimization algorithm, although it has low effectiveness than some other state-of-the-art optimization algorithms. In the random search algorithm, by inputting the dimensions of the input, the boundary of each input parameter, and the maximum iteration number, we can get the optimal geometry modification parameters set with the lowest total resistance based on the random search algorithm. Due to the relatively large time cost of the potential flow method, we set the maximum iteration number of 200 in this work.

Based on the random search algorithm, we executed the simulation-based design for the FCNN model, the MINN model, and the potential flow method. Figure 16(*a*) shows how the total resistance of the optimal hull form changes with the iteration number. As shown in Fig. 16(*a*), three methods acquired very similar total resistance values when iteration numbers reach 200, showing the design assistant ability of both the FCNN model and the MINN model. Due to the randomness of the random search algorithm, the final total resistance values obtained based on different methods are different. It is also worth noting that the MINN model reaches convergence earlier than those of the other two methods. From Fig. 16(*a*), we can see the similarities in the design process for all three methods. Figure 16(*b*) shows how the total resistance of the optimal hull form changes with calculation time, and one can find a significant difference among the three methods. For both the MINN model and the FCNN model, the random search algorithm can find a low-resistance hull within 100 s. However, for the potential flow method, a random search with 200 iterations will cost over 18,000 s—a total of 5 h.

From Fig. 16(*b*), one can find that the artificial intelligence-assisted hull shape design can help designers save much time. It is worth noting that the design sample used in the above calculations is simple, and for the actual ship design, the hull form configuration can be much more complex, the artificial intelligence-assisted design (AIAD) can save a tremendous amount of time. Thus, we conclude that either the MINN model or the FCNN model can assist the hull form design and shorten the design time and hence workload. Moreover, because the MINN model can predict the trend between the geometric modification parameters and the total resistance more precisely, the MINN model offers better reliability in the design process. The higher trust for the MINN model is also demonstrated in Table 10, which shows the difference between the total resistance of hull predicted by the FCNN model and the MINN model with the result predicted based on the potential flow method.

Table 10 Comparison of best hull actual resistance and predicted resistance

Model	Actual	Predicted	Error
MINN	42.81	40.43	5.79%
FCNN	36.70	39.19	6.78%

7 Conclusions

In this work, we have developed a MINN model to predict the total resistance of the ship hull designed based on the geometry modification parameters while avoiding inconsistent accuracies in evaluating the performance of various types of input parameters. In the AI-assisted design, we used the KRISO container ship (KCS) as the initial hull model template and simplified the geometry modification process using two categories of geometry modification parameters. By exploring the design space of the hull structure model, we randomly generated hull shape form samples and adopted the potential flow method to calculate the total resistance of all hull samples. The final MINN model was developed based on a series of tuning and training processes.

The final forecast accuracy verification and sensitivity similarity verification show that the architecture of the MINN model can predict hull performance with high accuracy and avoid the problem of different sensitivity caused by the different degrees of influence of the input parameters. In summary, we find that the application of the MINN architecture in the ship hull optimization shows remarkable success in applying AIAD in ship hull shape design.

Based on this study, we have concluded that the MINN model can predict the trend of ship hull performance with geometry modification parameters with high accuracy in ship hull shape design, providing designers with an efficient tool that immediately maps design requirements with structure configurations. On the other hand, whether or not the MINN model can perform better in trend predictions for other design cases still remain to be seen or to be studied, even though we believe that the MINN model may have the potential to be used in other design problems of different sensitivities when applied in other mathematical or engineering applications.

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Conflict of Interest

There are no conflicts of interest.

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Data Availability Statement

The data and information that support the findings of this article are freely available.[2] No data, models, or code were generated or used for this paper.

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